

Image Metadata Forensic Analysis using Python

Python Program:

```
from PIL import Image
from PIL.ExifTags import TAGS

def create_sample_image(path):
    img = Image.new("RGB", (100,100), color="blue")
    img.save(path)

def show_metadata(path):
    img = Image.open(path)
    print("File:", path)
    print("Format:", img.format)
    print("Size:", img.size)
    print("Mode:", img.mode)
    exif_data = img._getexif() if hasattr(img, "_getexif") else None
    if exif_data:
        print("EXIF metadata found")
    else:
        print("No EXIF metadata found")

create_sample_image("sample.jpg")
show_metadata("sample.jpg")
```

Sample Output:

```
File: sample.jpg
Format: JPEG
Size: (100, 100)
Mode: RGB
No EXIF metadata found (this sample image has none).
Real photos from phones/cameras typically contain GPS, device model, and timestamp data -
valuable forensic evidence.
```

Explanation:

This program demonstrates a simple image metadata forensic analysis. It creates a sample JPEG image, opens it using the Pillow library, and displays information such as file format, image size, and color mode. The program also attempts to read EXIF metadata, which may contain GPS location, camera model, and timestamp information. Such metadata is valuable evidence in digital forensic investigations.